Engine Model(s): PT6A

72-50-01

COMPRESSOR TURBINE STATOR - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03 which is 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-367	Compound, Lapping
PWC05-368	Compound, Lapping
PWC11-010	Alcohol, Methyl
PWC11-012	Acetone
PWC11-014	Alcohol, Isopropyl
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30478	Spacer	
PWC30863	Puller	Obsolete - Replaced by PWC32822
PWC32209	Adapter	
PWC32380	Gage, Shroud	
PWC32396-400	Jackscrew, Accessory Gearbox	Replaces PWC42034
PWC32528	Wheel, Grinding	

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Tool No.	Name	Application
PWC32822	Puller	Obsolete - Replaced by PWC71559
PWC37917	Dresser, Diamond, Grinding Wheel Dressing	
PWC37918	Grinder, Compressor Turbine Shroud	
PWC42034	Jackscrew, Accessory Gearbox	Replaced by PWC32396-400
PWC64241-1 or -5	Guidepin	
PWC71559	Puller	Alternate to PWC32822

4. Fixtures, Equipment and Supplier Tools

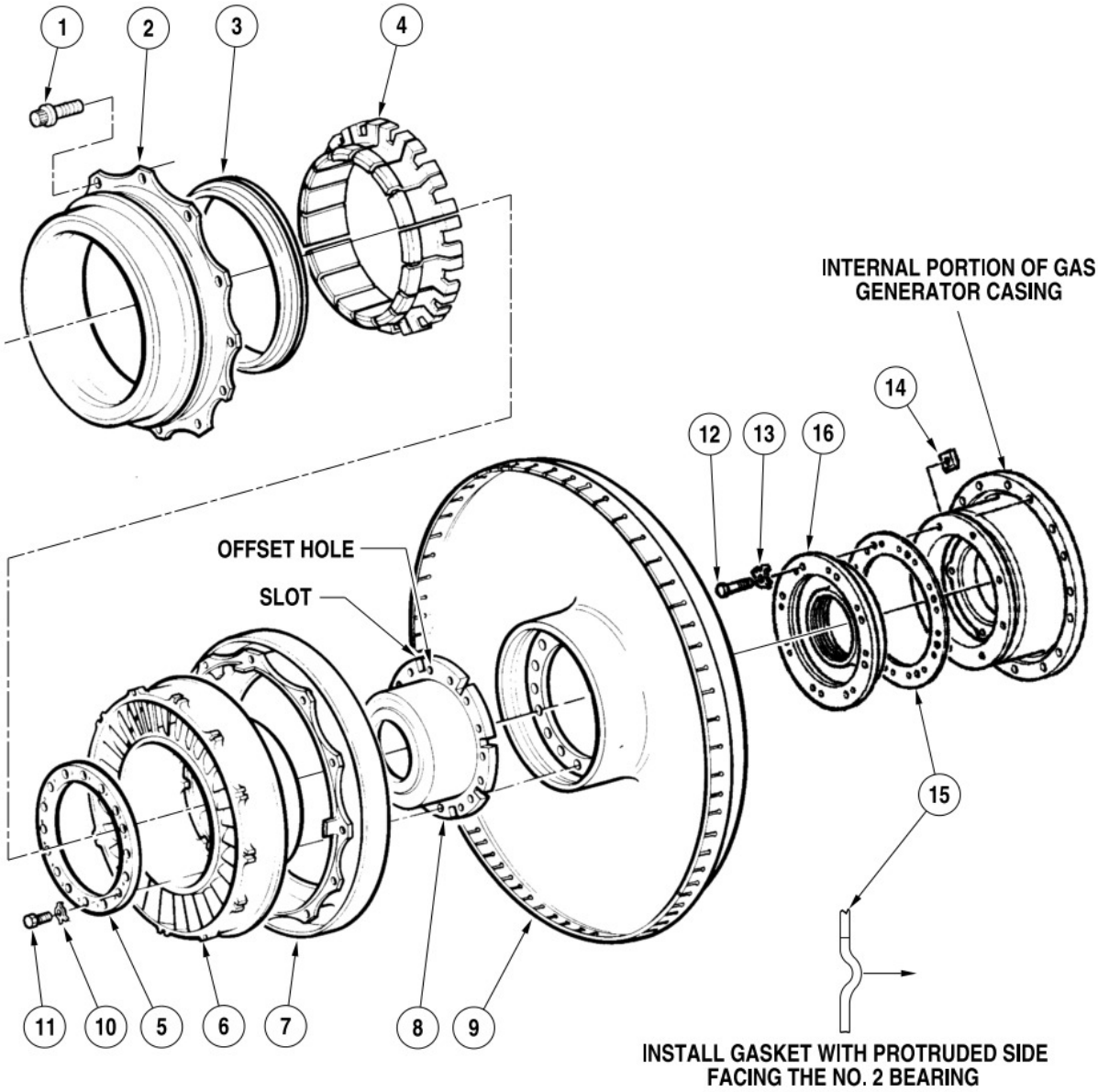
Not Applicable

5. Removal/Installation (Heavy Maintenance Only)

A. Removal of Compressor Turbine Stator (Ref. Fig. 201)

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Figure 201 Removal/Installation of Compressor Turbine Stator Assembly



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1. Silver-plated Bolt
2. Shroud Housing
3. Heatshield
4. Shroud Segments (Post-SB1158)
5. Lockplate
6. Vane Ring
7. Small Exit Duct
8. No. 2 Bearing Cover
9. Large Exit Duct
10. Tabwasher
11. Bolt
12. Bolt
13. Keywasher
14. Nut
15. Gasket
16. No. 2 Bearing Stator Air Seal

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(1) Remove the power section from the gas generator assembly (Ref. 72-00-00).

(2) Install the power section in a stand (Ref. 72-00-00).

(3) Remove the compressor turbine disk and blade assembly (Ref. 72-50-02).

(4) Remove the combustion chamber liner (Ref. 72-40-01).

(5) Remove bolts and tabwashers securing the lockplate (5), compressor turbine vane ring (6), shroud housing assembly (2) and small exit duct (7) to the gas generator case.

NOTE: Before moving the compressor turbine vane ring, mark the location of the vane ring relative to the no. 2 bearing cover flange using an approved marker (Ref. 70-00-00).

(6) Remove the lockplate (5), compressor turbine vane ring (6), shroud housing (2) and small exit duct (7) assembly from the large exit duct and gas generator case.

CAUTION: DELETED.

(7) Remove the No. 2 bearing cover (8) with the puller (PWC30863) or (PWC71559).

NOTE: Post-SB1231 engines have the four threaded holes. If necessary, remove the cover with jackscrews (PWC42034) or (PWC32396-400) in these holes. Material thickness of cover flange is such that it is necessary to carefully tighten the jackscrews to prevent damage to threads.

(8) Remove compressor turbine stator air seal (16) and gasket (15). Discard the gasket (15).

(9) Remove the large exit duct (9) from the gas generator case.

NOTE: For Pre-SB1178 Engines: Apply hand pressure evenly on each side of centerbore of large exit duct, and at the same time rotate duct in a counterclockwise direction for approximately 23 degrees to disengage scallops from support brackets on the gas generator case.

(10) Using an approved marker (Ref. 70-00-00, Marking of Parts), index mark the compressor vane ring (6), small exit duct (7), heatshield (3, Pre-SB1305) and shroud housing (2) to correctly align mating surfaces on reassembly.

(11) Remove bolts (1) which secure the vane ring (6) and shroud housing (2) to the small exit duct (7).

(12) Separate the vane ring (6) from the shroud housing (2) and small exit duct (7).

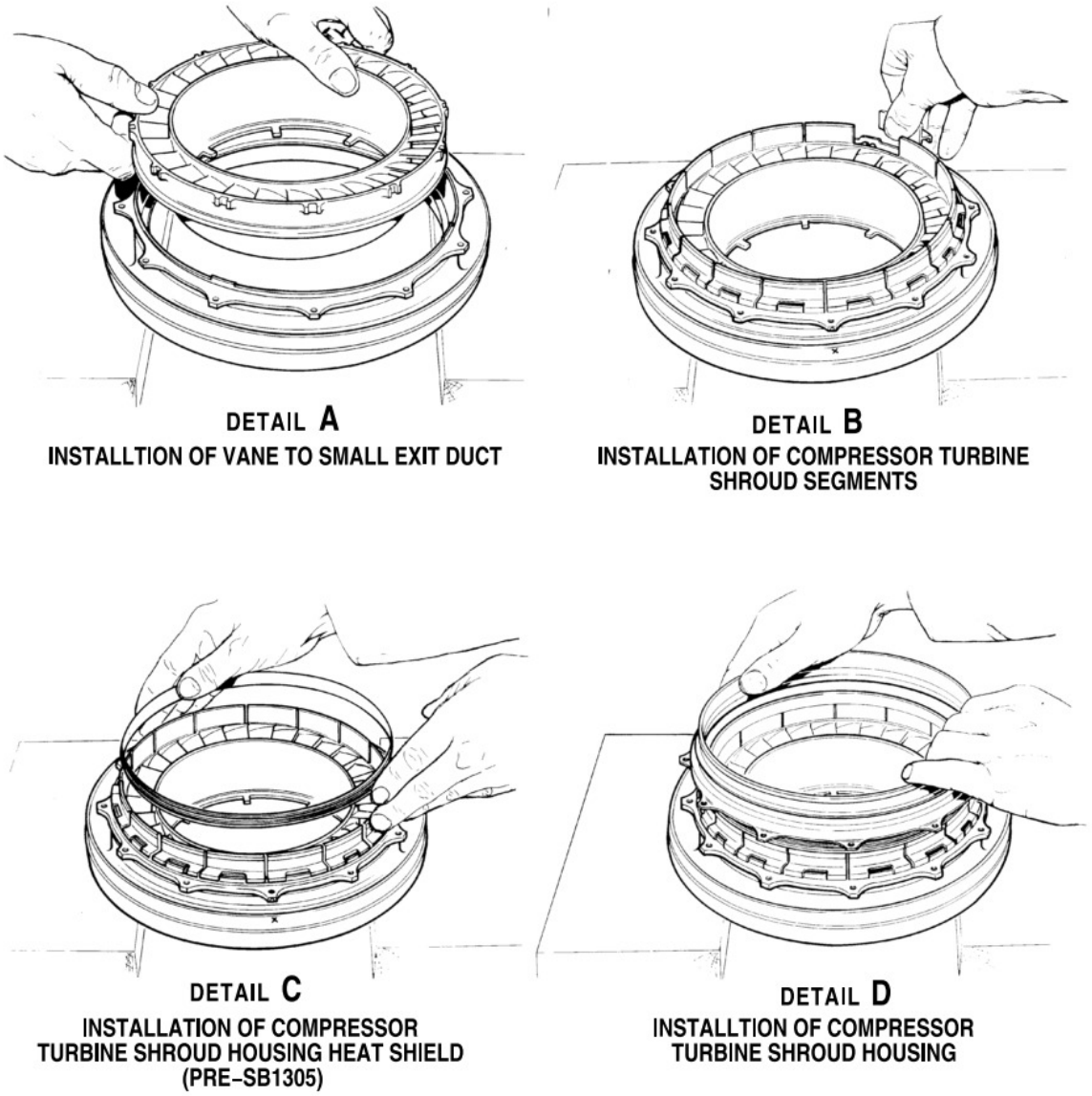
(13) Remove shroud segments and heatshield (3, Pre-SB1305) from shroud housing.

NOTE: Unless shroud segments are to be replaced, the assembly should not be disassembled or disturbed if avoidable.

B. Installation of Compressor Turbine Stator (Ref. Figs. 201, 202 and 203)

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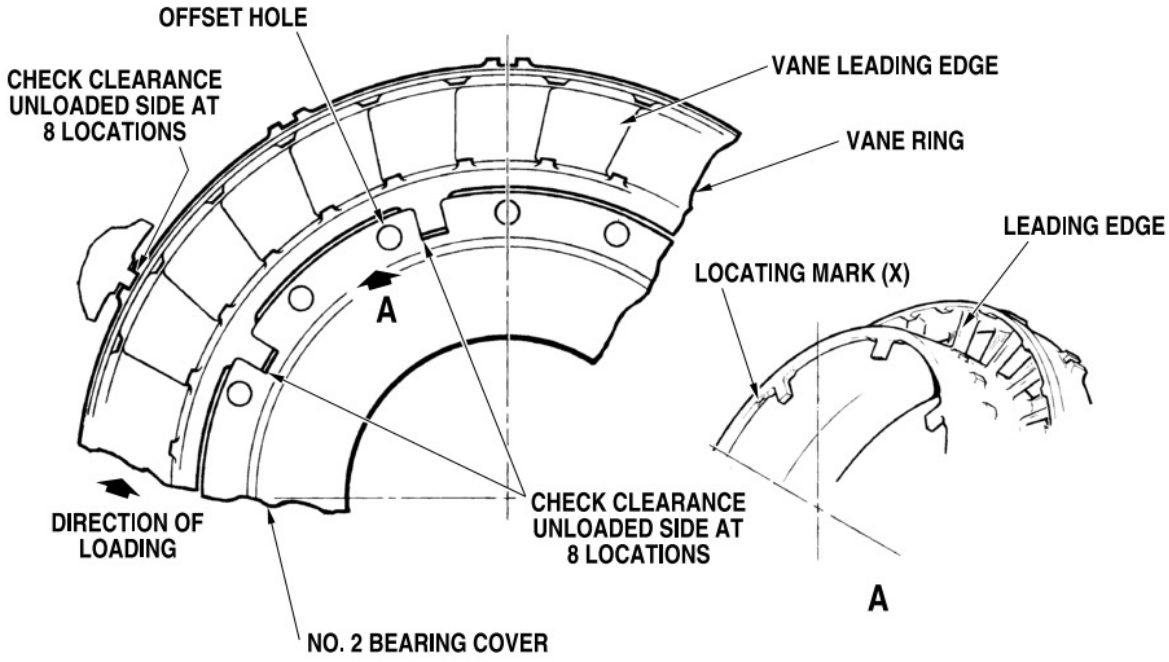
Figure 202 Compressor Turbine Vane Ring and Shroud Housing Assembly Sequence



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Figure 203 Small Exit Duct/Vane Ring Clearance Check



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(1) Place small exit duct (7, Fig. 201) on wooden blocks, flange uppermost, and install compressor turbine vane ring (6) in ID of duct. Make sure slots in ring engage with corresponding lugs on duct and vane ring is correctly seated and centralized. Index (rotate) vane ring until best fit is obtained.

(2) Check clearance between slots in vane ring and lugs on duct, take measurements on unloaded side of slots at all locations. Clearance should be between 0.002 and 0.005 inch.

(3) If insufficient gap exists at any location, appropriate slots in vane ring may be lightly stoned until acceptable clearance of 0.002 to 0.005 inch is obtained at all locations.

NOTE: If side play is found in excess of 0.002 inch, vane ring and small exit duct must be reworked at an approved overhaul facility. Excessive clearance can displace true center of shroud housing and ground shroud segments from the running center of compressor turbine disk. This condition may cause a blade tip rub on one side and an excessive tip clearance on diametrically opposite sides.

(4) Install shroud segments on vane ring:

(a) Starting at most convenient point, position segments of appropriate class in sequence on vane ring (Ref. Fig. 202, Detail B).

(b) Clean ID of segments with alcohol (PWC11-010)/(PWC11-014) or acetone (PWC11-012), then apply masking tape to cleaned area to prevent segments from falling out of position during subsequent handling.

(c) Locate heatshield (Pre-SB1305) over shroud segments (Ref. Fig. 202, Detail C). Align index marks on vane ring and heatshield.

(5) Locate compressor turbine shroud housing over vane ring and small exit duct, align index marks on housing and duct, then press housing into position.

(6) Remove masking tape, if used at assembly, and thoroughly clean off all traces of adhesive with alcohol or acetone.

(7) Secure shroud housing to small exit duct with bolts (1, Fig. 201). Torque bolts 20 to 25 lb.in. and lockwire in pairs.

(8) Inspect ID of vane ring (adjacent to lugs) for index mark (Ref. Fig. 203). If mark is legible, align mark with offset hole in No. 2 bearing cover (8, Fig. 201) then proceed with assembly at Step (10) .

(9) If the index mark is not legible, or if a new vane ring is being installed (Ref. Fig. 203):

(a) Install assembly on No. 2 bearing cover.

(b) Check clearance between lugs on vane ring and slots in No. 2 bearing cover, taking measurements on unloaded side of slot at all eight locations. A 0.001 inch feeler gage should enter gaps between lugs and slots, but a 0.003 inch feeler gage must not.

NOTE: If the clearance or side play is found to be excessive, it must be corrected by rework at an overhaul facility. Excessive clearance can displace the true center of shroud housing and the ground shroud segments from the running center of compressor turbine disk. This condition may cause a blade tip rub on one side

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and excessive tip clearance on the diametrically opposite side.

(c) If insufficient gap exists at any location, the appropriate lug(s) may be lightly stoned until correct clearance of 0.001 to 0.002 inch is obtained at all eight locations. Minimum lug width after rework to be not less than 0.245 inch.

(d) Vibropeen index (X) mark on ID of vane ring adjacent to offset hole in No. 2 bearing cover (Ref. Fig. 203).

(10) Make sure the mating surfaces of vane ring, shroud housing, No. 2 bearing cover and large exit duct are clean.

(11) Locate large exit duct (9, Fig. 201) on flange of gas generator case, and align offset holes.

NOTE: For Pre-SB1178 Engines: Locate exit duct in gas generator case with offset hole positioned approximately 23 degrees counterclockwise. Apply hand pressure to centerbore of duct while rotating clockwise to engage scallops on rear face of duct with brackets in case. Align offset hole in duct with corresponding hole in case.

(12) Locate No. 2 bearing cover (8) on large exit duct (9) and align offset holes.

NOTE: Due to relative tolerances, tightness may be encountered when fitting cover on gas generator case. When such conditions prevail, the use of a soft-faced hammer is permissible.

(13) Secure No. 2 bearing cover:

(a) Install a new gasket (15) (with the protruded side towards the No. 2 bearing) and No. 2 bearing stator air seal (16) on flange of gas generator case. Install two guide pins (PWC64241-1) or (PWC64241-5) into gas generator case.

(b) Secure No. 2 bearing stator air seal (16) with keywashers (13) bolts (12) and nuts (14). Remove the guide pins and install the remaining bolts. Torque the bolts 32 to 36 lb.in. and lock keywashers.

NOTE: Depending on SB standard of gas generator case, Dee-nuts (14) may be present.

(14) Locate vane ring and shroud housing assembly on to the No. 2 bearing cover. Align index mark on vane ring with offset hole in bearing cover, then engage lugs on vane ring with slots in cover. Install lockplate (5) and secure with tabwashers (10) and bolts (11). Tighten bolts and torque 36 to 40 lb.in. Check that vane ring and shroud housing assembly can be moved slightly by hand (REF. NO. 489). Lock tabwashers. Total side play of assembly should not exceed 0.003 inch.

(15) Install compressor turbine disk and blade assembly (Ref. 72-50-02).

(16) Check compressor turbine tip clearance (Ref. 72-50-02).

(17) Install combustion chamber liner (Ref. 72-40-01).

(18) Install power section (Ref. 72-00-00).

6. Inspection/Check

A. Compressor Turbine Vane Ring (Ref. Table 201 and Fig. 205)

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(1) Inspect turbine vane ring.

(2) Inspect turbine shroud housing and shroud segments:

(a) Inspect OD of shroud housing for wear and grooving in contact area with interstage sealing ring. If wear is uneven and conducive to gas leakage, return housing to an approved overhaul facility for repair.

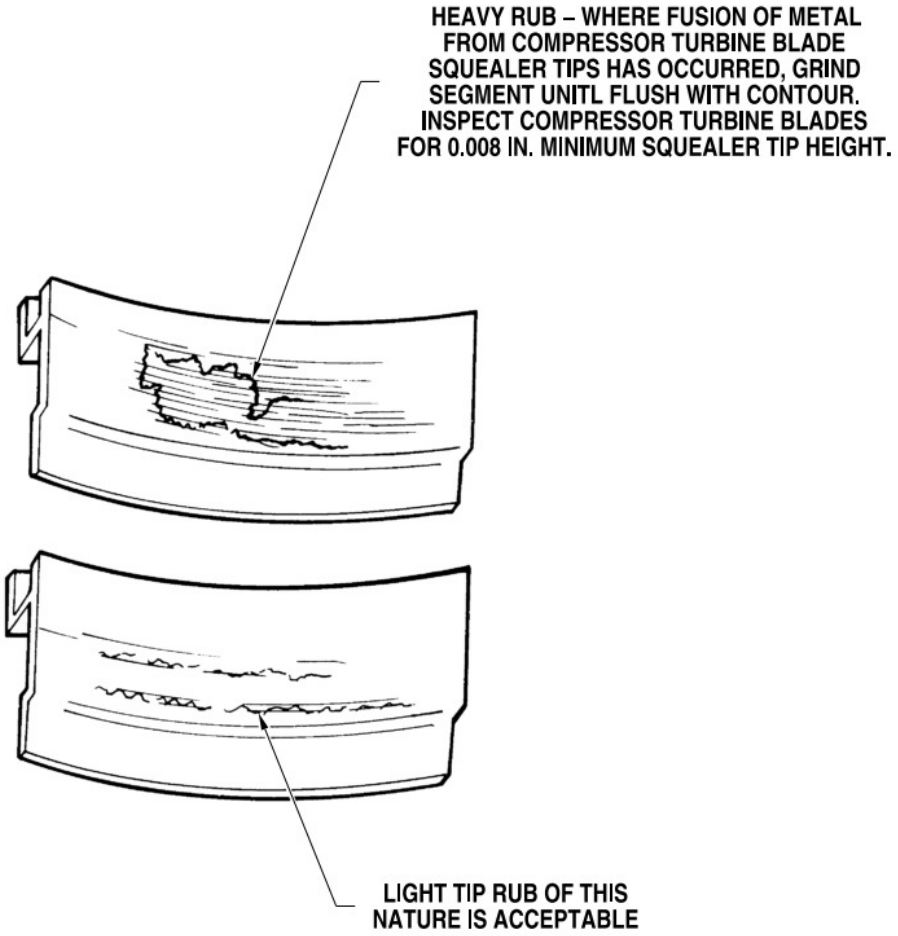
(b) Inspect shroud housing (Post-SB1305) for cracks in shroud segment retaining lip. Cracks not exceeding 0.150 in. in length and separated by a minimum of 1.00 in. of sound material are acceptable. Shroud housings cracked in excess of limits are unserviceable; ship to an approved overhaul facility for possible repair.

(c) Inspect compressor turbine shroud segments for metal build-up and cracks (Ref. Fig. 204).

NOTE: To maintain concentricity and accuracy during machine grinding, No. 2 bearing cover should accompany compressor turbine hot section hardware returned to a repair facility.

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Figure 204 Metal build-up and Cracks on Shroud Segments



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Table 201 Compressor Turbine Stator Inspection

Inspection	Max. Serviceable	Max. Repairable	Corrective Action
COATING LOSS			
Vanes	Not critical without erosion.	**	*
Outer ring	Not critical without erosion.	**	*
Inner ring	Not critical without erosion.	**	*
CRACKS			
Vane trailing edges	Open cracks up to 0.500 inch long are acceptable on any number of vanes provided their progression will not result in loss of vane materials.	**	*
CAUTION: OUTER RING CRACKS ARE ACCEPTABLE ONLY IF SLOT-TO-LUG CLEARANCE CAN BE OBTAINED AT ASSEMBLY.			
Inner ring	Up to ten through cracks are acceptable provided they are not convergent and/or do not display signs of air leakage.	**	*
Outer ring	Three cracks across the length of the outer ring are acceptable. Outer ring cracks that extend into vanes are acceptable. Multiple hairline cracks in, but not across, the ring are acceptable. Convergent cracks are not acceptable.	**	*
EROSION AND BURNING			
Vanes	0.005 inch deep on any airfoil surface on any number of vanes. When compressor turbine vane ring inspection reveals that vane trailing edge burning (Ref. Fig. 205) has resulted in loss of airfoil area of 0.125 sq.in. or more (i.e. average length multiplied by average width; eg. length 0.350 x width 0.350 = 0.1225 sq.in.), on any one vane, and burned area can be seen from leading edge of vane ring when viewed parallel to engine longitudinal axis, the entire set of compressor turbine blades must be scrapped (Ref. 72-50-02).	**	*
Outer and Inner	Any amount.		

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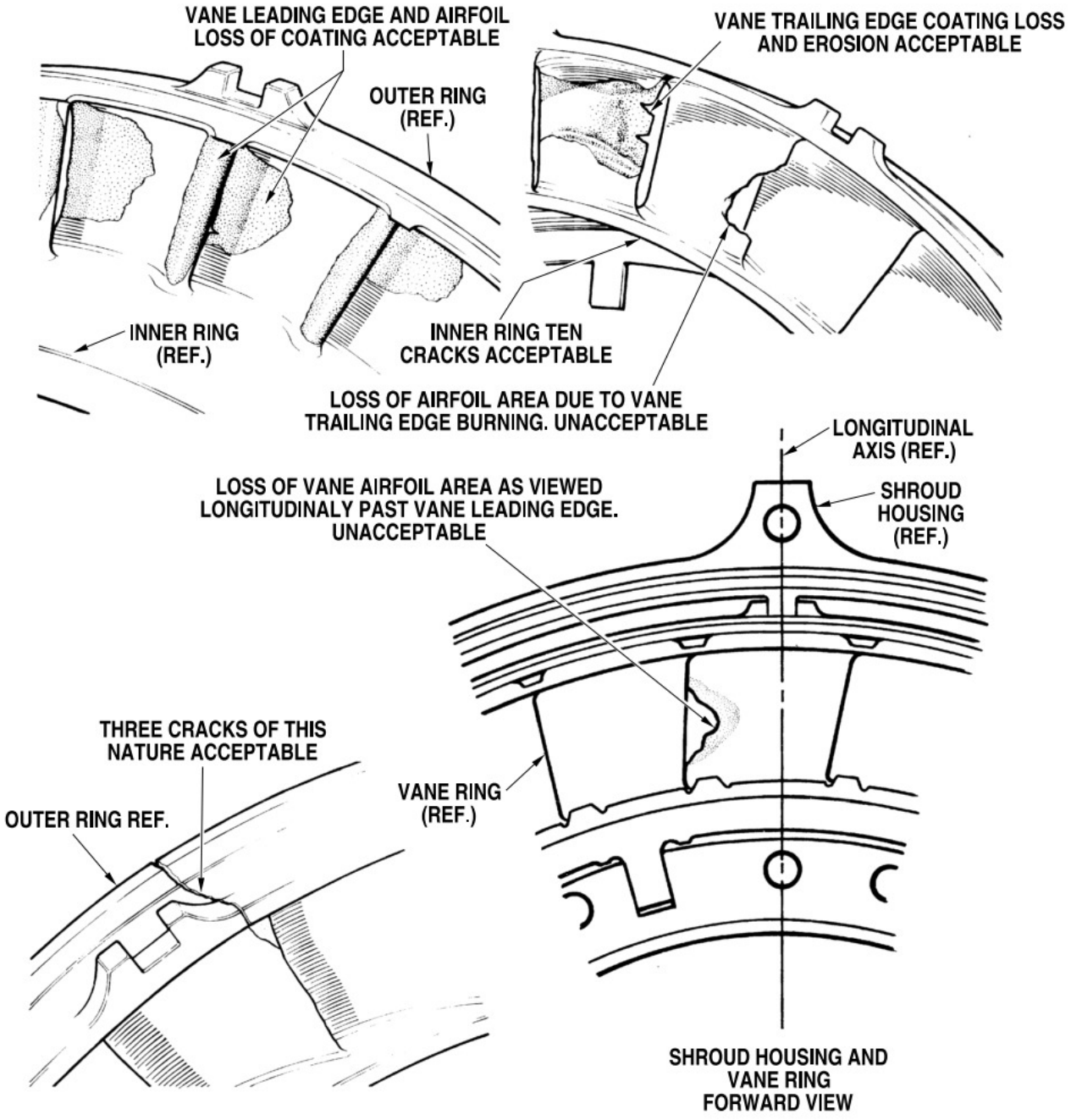
Inspection rings	**Max. Serviceable**	**Max. Repairable**	**Corrective Action**
CONTOUR	Vane airfoil contour distortion or surface irregularities may degrade engine performance. A cracked outer ring may incur geometric location errors in the slots which effect small exit duct fit and therefore shroud segment concentricity. These effects must be considered in determining the serviceability of the vane ring.		
GENERAL NOTES			
If distressed vanes are encountered, check fuel nozzles functioning.			
Acceptance criteria is based on structural integrity considerations and apply only if engine performance was satisfactory prior to an HSI.			
Replace stator assembly if its condition appears to be the cause for loss of performance. If stator assembly must be replaced, an assembly of same class or ± 0.03 in^2 or ± 0.12 class (REF. S.I.L. No. GEN PT6A 020).			
* A stator exhibiting defects in excess of Maximum Serviceable limits must be replaced.			
**Stator considered unacceptable for continued service may be sent to P&WC or following vendors for possible repair. Pratt & Whitney Canada 7007 de la Savane St. Hubert, Quebec Canada J3Y 3X7 Attention: Receiving Area No. 9 Vac-Hyd, Dallas, Blade and Vane Division 1177 Great Southwest Parkway Grand Prairie, TX 75050 USA Howmet Turbine Components Corp. 555 Benston Rd. Whitehall, MI 49461 USA Attention: Thermatech Division Sifco Turbine Component 2430 Winnetka Avenue North Minneapolis, MN 55427 USA			

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Inspection	Max. Serviceable	Max. Repairable	Corrective Action
Classification: When compressor turbine vane ring is to be replaced, a vane ring of the same class, or within ± 0.03 in^2 or ± 0.12 class variation of the class from the last test cell performance run, is recommended for installation. If a vane ring class is unknown, refer to the last test cell performance run (log book) or contact your local P&WC Customer Support Representative.			

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Figure 205 Compressor Turbine Vane Ring Inspection (Typical)



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B. Small and Large Exit Ducts

(1) Inspect the small exit duct for fretting wear at the duct flange which contacts the combustion chamber liner. Wear up to 0.010 inch is acceptable. If wear is in excess of 0.010 inch, the duct should be returned to an approved overhaul facility for possible repair.

(2) Radial cracks up to 1.00 inch long in the small exit duct outer wall are acceptable provided they are stop drilled using a 1/16 (0.0625) inch drill.

(3) Inspect the large exit duct for fretting wear along the area which contacts the combustion chamber liner. Wear up to 50 percent of the material thickness is acceptable. If the wear is in excess of 50 percent, the duct should be returned to an approved overhaul facility for possible repair.

(4) Cracks up to 1.00 inch long in the large exit duct inner wall are acceptable provided they are stop-drilled using a 1/16 (0.0625) inch drill.

(5) Coating loss of any amount is acceptable on both the small and large ducts provided there is not evidence of burning or erosion of the parent metal.

7. Approved Repairs (Heavy Maintenance Only)

A. Grinding Compressor Turbine Shroud Segments

(1) General

(a) To maintain concentricity and uniform turbine blade tip to shroud segment clearance, grinding should be carried out with all components, especially No. 2 bearing cover, fitted to the engine.

(b) Replacement of the compressor turbine shroud segments may entail localized grinding to remove high spots and/or any slight eccentricity of the segments to obtain the correct turbine blade tip clearance. If the tip clearance exceeds the specified limit (REF. NO. 318, Fits and Clearances, Chap. 72-50-01) after the removal of the high spots and/or eccentricity, select the next class (thicker) of segments (Ref. Table 202). Install (Ref. Para. 7.) and grind as necessary.

NOTE: Pre-assembled compressor turbine vane ring with shroud housing and bladed compressor disk procured as repaired and/or replacement assemblies, are usually machine ground to provide the minimum compressor turbine blade tip to shroud segment clearance. Upon installation, a final grind may be necessary to make sure the prescribed clearances are maintained.

Table 202 Compressor Turbine Shroud Segment Classification

Pre-machined Turbine Bladed Disk OD (in)	Recommended Class	Shroud Segment Dim. A (in.)		Grind Shroud Segment ID (in.)	
				Min.	Max.
8.556 max.	Class 1	0.042 - 0.043		8.575	8.581

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Pre-machined Turbine Bladed Disk OD (in)	Recommended Class	Shroud Segment Dim. A (in.)	Grind Shroud Segment ID (in.)	
			Min.	Max.
8.550	Class 2	0.045 - 0.046	8.569	8.575
8.544	Class 3	0.048 - 0.049	8.563	8.569
8.538	Class 4	0.051 - 0.052	8.557	8.563
8.532	Class 5	0.054 - 0.055	8.551	8.557
8.526	Class 6	0.057 - 0.058	8.545	8.551
8.520	Class 7	0.060 - 0.061	8.539	8.545
8.512 min.	Class 8	0.069 - 0.072	8.521	8.527
	Class 9	0.079 - 0.083	8.501	8.507

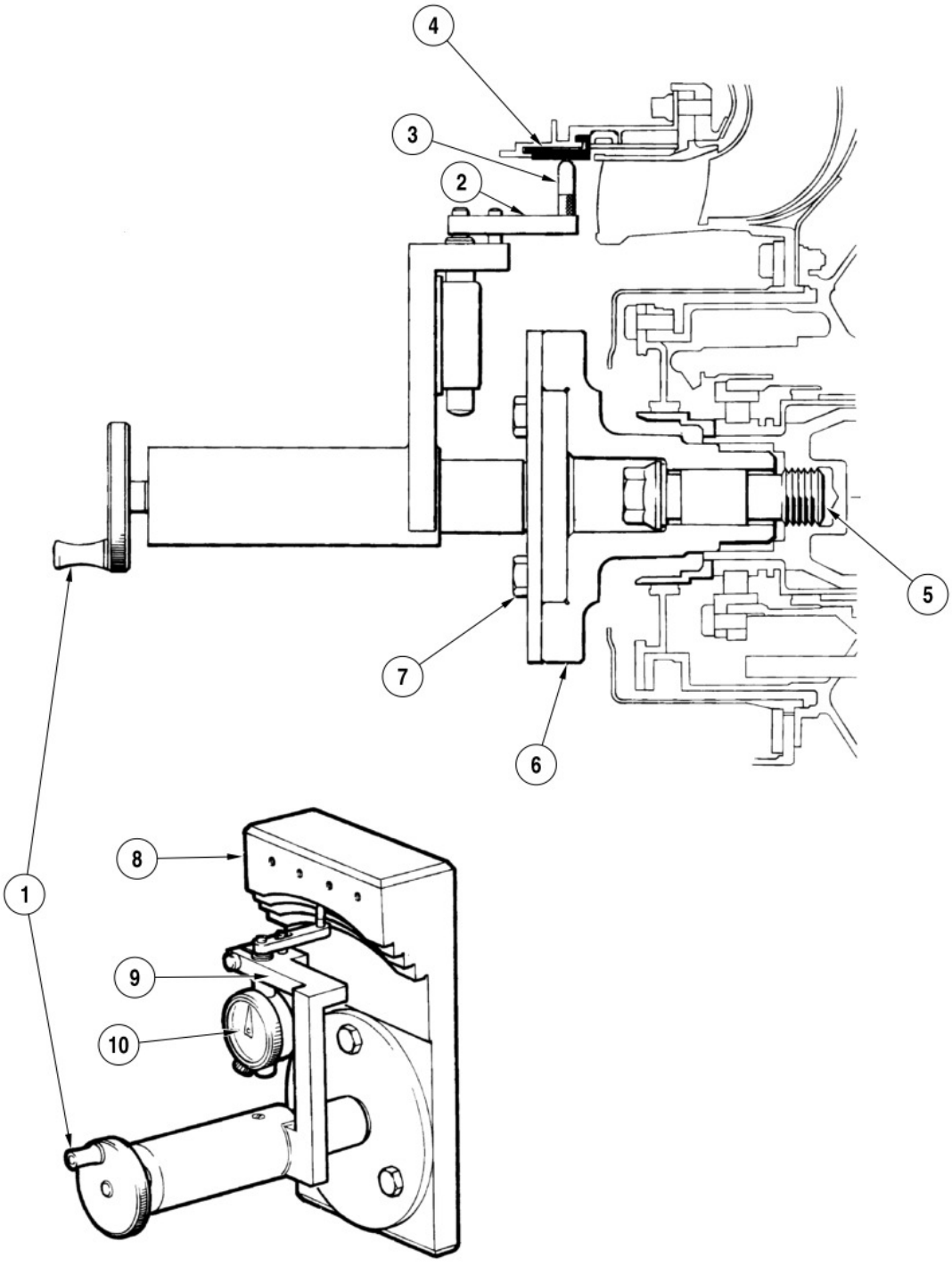
NOTE: 1. If the blade tip clearance exceeds the specified limits after removing the eccentricity of the shroud segments, select the next class (thicker) of segments and grind as necessary.

NOTE: 2. Minimum grind to remove ovality only. Minimum thickness of shroud segment after grinding on assembly is 0.055 in. Refer to Figure 206 for details of Dimension 'A'.

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Figure 206 Compressor Turbine Shroud Segments - Blade Tip Clearance Check

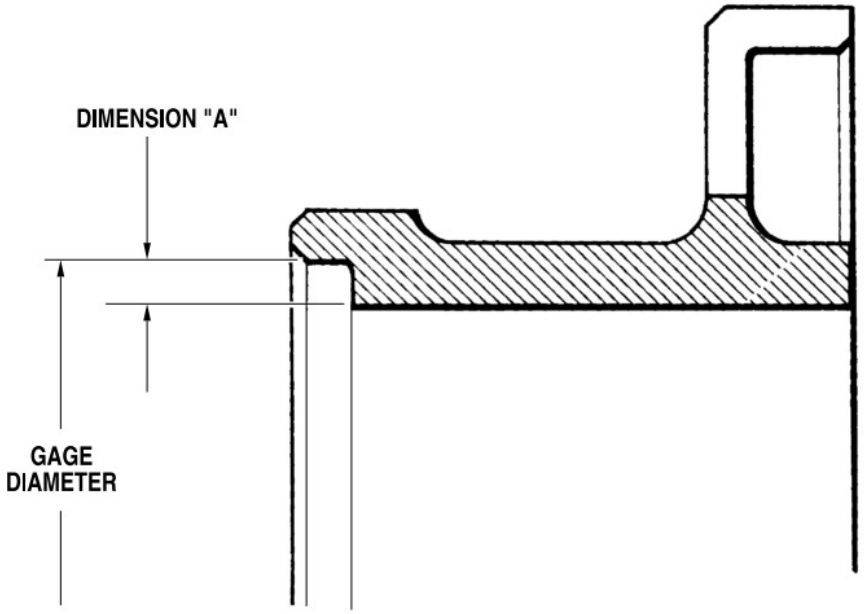
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(SHEET 2 OF 2)



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1. Axial Adjuster
2. Tip
3. Contact Point
4. Shroud Segment
5. Bolt
6. Adapter (PWC32209)
7. Bolt (3 req'd)
8. Gage Master
9. Gage Body (PWC32380)
10. Dial Indicator

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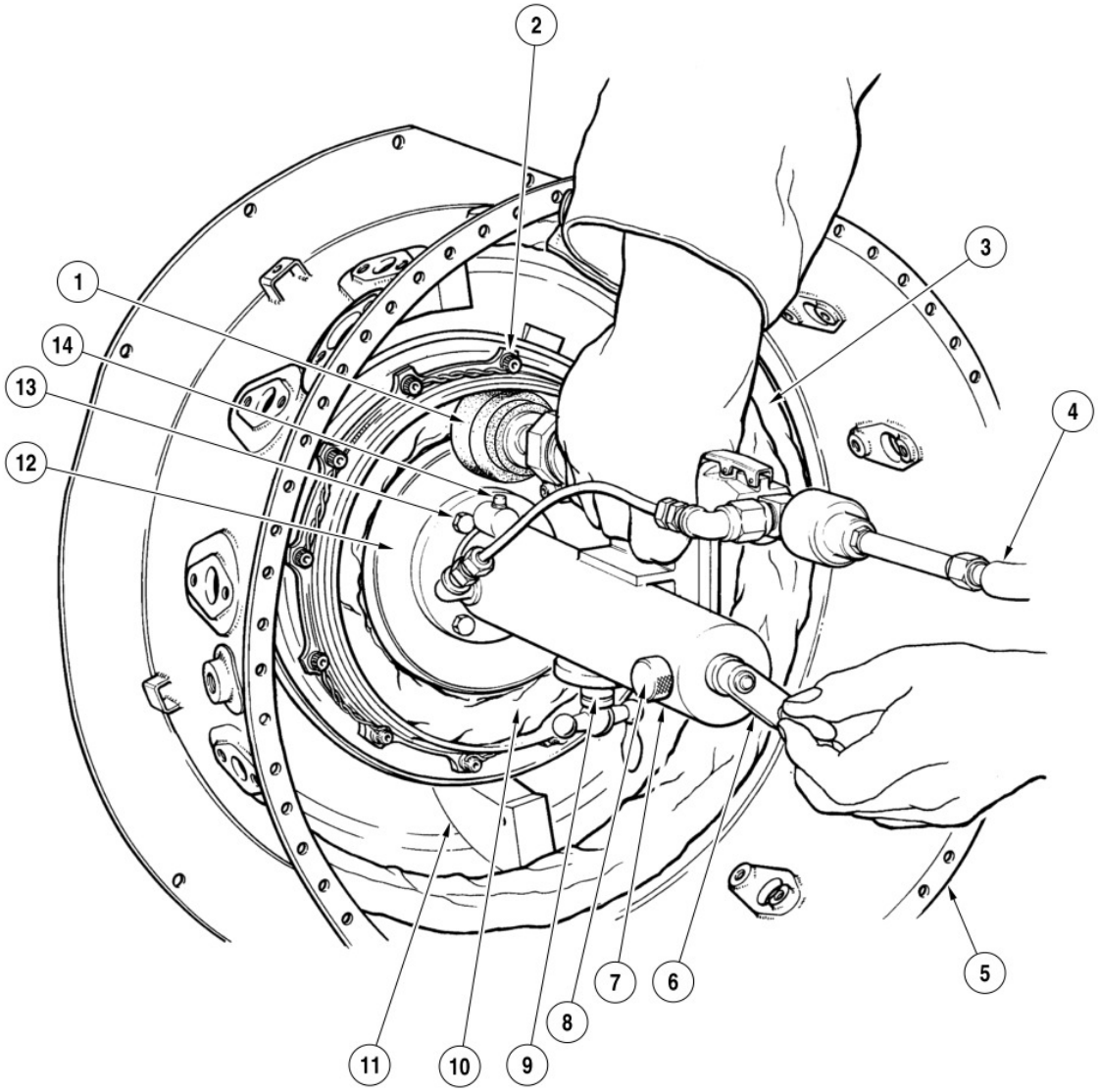
(2) Determine Shroud Segment Concentricity and Diameter (Ref. Fig. 206).

(a) Install adapter (6, PWC32209) on compressor front stubshaft and secure with bolt (5).

(b) Install four spacers (11, Fig. 207, PWC30478) under the small exit duct. Set the spacers approximately 90 degrees apart.

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Figure 207 Compressor Turbine Shroud Segment Grinding



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1. Grinding Wheel (PWC32528)
2. Bolt (Shroud Housing)
3. Blanking Material (Exit Duct)
4. Air Hose
5. Gas Generator Case
6. Axial Vernier Adjust
7. Grinder (PWC37918)
8. Axial Adjustment Lockscrew
9. Radial Vernier Adjust
10. Blanking Material (Stator Assembly)
11. Spacer (PWC30478)
12. Blanking Disk
13. Bolt
14. Diamond Dresser (PWC37917)

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(c) Select the contact point (3, Fig. 206) for the radius to be measured and fasten to the radius gage tip (2).

NOTE: The radius gage and gage master are common to all PT6 engines. The gage is supplied with four contact points of different lengths, while the gage master has four "steps" of different radii. These variables allow the gage and master to be used on all PT6 engine models.

(d) Install the radius gage (9) on the adapter (6) with bolts (7). Tighten the bolts 20 to 30 lb.in. Check for the correct mounting by trying to insert a 0.001 inch shim stock between the gage and adapter.

(e) Position the contact point (3) at the center of the segment (4) using the adjuster (1).

(f) Take four readings, 90 degrees apart around the shroud segments. If the shroud segments are not concentric with the rotor shaft, check the correct positioning of the spacers (11, Fig. 207).

(g) Remove the radius gage from the adapter (6, Fig. 206) and install in the gage master (8). Set the dial indicator (10) to zero.

(h) Remove the radius gage from the master (8). Make sure the dial indicator setting is not disturbed.

(i) Install the radius gage on the adapter (6) with bolts (7). Tighten the bolts 20 to 30 lb.in. Check that a 0.001 inch shim stock will not fit between the gage and adapter.

(j) Rotate the gage and record the high and low readings. Adjust the gage in and out to determine the segment taper. A low reading and/or segment taper will determine whether the segments are suitable for further use or if replacement is required.

(3) Calculate the grinding or replacement of the shroud segments as shown in following typical examples:

Where: A = True radius as stamped on gage master

B = Radius of bladed disk, $^1/_2$ OD, as measured with micrometer

C = Dial indicator reading on shroud segments

D = Radius of shroud segments

E = Existing blade tip clearance, before grinding

EXAMPLE 1:

Where A = 4.282 inches

B = 4.266 inches

C = 0.013 inch clockwise reading on the minus side of the gage zero datum

D = A minus C = 4.282 inches minus 0.013 inch = 4.269 inches

E = D minus B = 4.269 inches minus 4.266 inches = 0.003 inch

(4) In this example, 0.010 inch must be ground from the existing shroud segments to provide a

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minimum blade tip clearance of 0.013 inch.

EXAMPLE 2:

Where A = 4.282 inches

B = 4.266 inches

C = 0.012 inch counterclockwise reading on the plus side of the gage zero datum

D = A plus C = 4.282 inches plus 0.012 inch = 4.294 inches

E = D minus B = 4.294 inches minus 4.266 inches = 0.028 inch

(5) In this example, the existing blade tip clearance exceeds the maximum permissible limit of 0.016 inch.

EXAMPLE 3:

Where A = 4.282 inches

B = 4.266 inches

C = 0.002 inch, clockwise reading on the minus side of the gage zero datum.

D = A minus C = 4.282 inch minus 0.002 inch = 4.2800 inches

E = D minus B = 4.280 inches minus 4.266 inches = 0.014 inch

(a) In this example, the existing blade radial tip clearance is within the limits and therefore no grinding of the shroud segments is required, except for the removal of local high spots.

(6) Grinding (Ref. Fig. 207)

(a) Using suitable material (3, 10 and 12), blank off the cavities around the combustion chamber liner and compressor turbine vane ring.

(b) Install the dresser (PWC37917) and grinding wheel (1, PWC32528) on the grinder (7) and install the grinder (7) on the adapter with bolts (13). Tighten the bolts 20 to 30 lb.in. Check that a 0.001 inch shim stock will not fit between the grinder and adapter to be sure of correct mounting.

NOTE: Dressing of the wheel may be done before or after the installation of the grinder on the adapter, or during the grinding operation.

(c) Position the wheel (1) in line with and slightly overlapping the segments, using the axial adjustment (6). Allow the wheel to touch the vane ring. Back off the adjustment (6) one turn and lock using the screw (8).

(d) Rotate the grinder around the segments, adjusting the radial adjustment (9) until the wheel (1) contacts the high spots. Back off the radial adjuster one turn.

(e) Connect the grinder air line (4) to the shop air of 90 psig with a flow rate of 36 SCFM.

NOTE: Where pronounced high spots exist, rotate the shroud grinder, back and forth, through an arc of 40 to 50 degrees to reduce the high spots before carrying out the normal 360 degree grinding.

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CAUTION: ROTATE THE GRINDER SLOWLY AND AVOID REMOVING AN EXCESSIVE AMOUNT OF MATERIAL IN ONE PASS.

(f) Operate the grinder. Adjust the wheel (1) with the radial adjustment (9) until the wheel contacts the segments, and grind the upper half of the segment ID to the required dimension.

NOTE: When grinding the upper half of the segments, do not load the grinder to compensate for any compressor rotor bearing play. Support the grinder when grinding the lower segment half to prevent taper grinding.

(g) Loosen the radial adjuster until the grinding wheel contacts the high spot on lower segments. Continue grinding, using complete revolutions, until the required dimensions are obtained.

(h) Disconnect the air hose (4) and remove the grinder from the adapter.

(i) Install the radius gage and recheck the shroud segment diameter. Check the segment taper by running the contact point across the width of segments.

(j) Remove the radius gage and return to the gage master. Remove the adapter.

(k) Remove all grinding residue from gas generator case with a suitable suction cleaner.

(l) Remove all blanking material and repeat step (k).

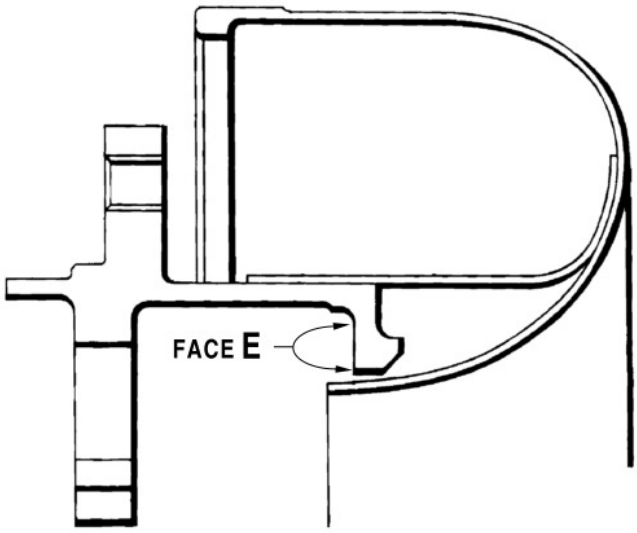
(m) Remove all masking adhesive residue with solvent (PWC11-027) or (PWC11-031).

(n) If the grinding has produced a lip with a sharp edge on the inner edge of the segment, or is raised from the ground surface by more than 0.005 inch, remove by local grinding with a hand held rotary tool, grinding wheel attachment and flexible drive, or similar equipment.

B. Restoration of Hot Section Sealing Surfaces (Ref. Fig. 208, 209 and 210)

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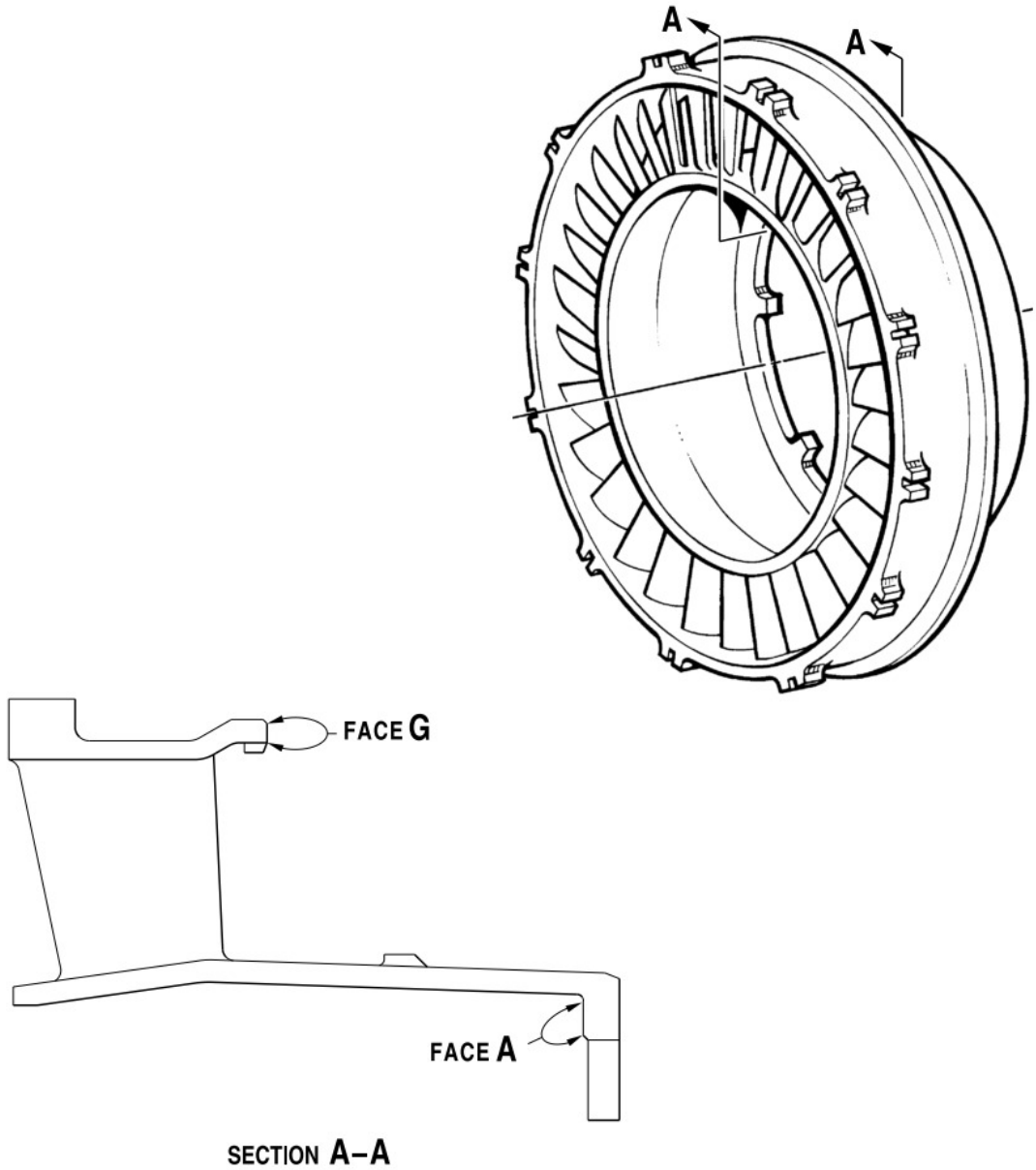
Figure 208 Small Exit Duct - Repair



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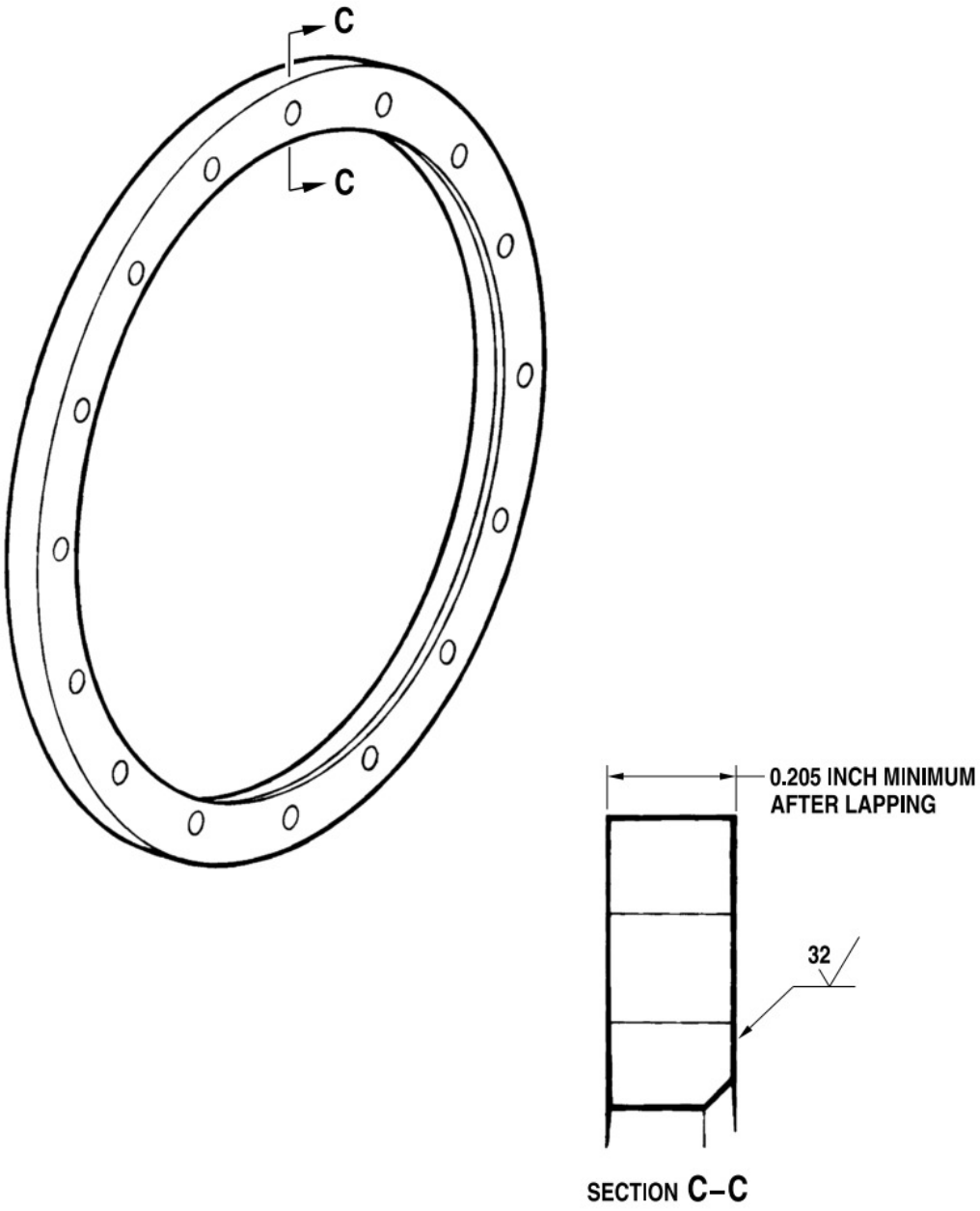
Figure 209 Compressor Turbine Vane Ring - Repair



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Figure 210 Vane Ring Lockplate - Repair



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(1) Gas tightness and sealing of the hot section surfaces depends on the surface finish, flatness, degree of oxidation and fretting.

(2) Lap the surface with lapping compound (PWC05-367) or (PWC05-368) to get a surface finish better than 32 AA and surface waviness of 0.0008 inch to get 100% contact for gas tightness.

(3) Lap face E (Ref. Fig. 208) of the small exit duct until 100% of the lapped finish of the contacting area is achieved.

(4) Lap face A and G (Ref. Fig. 209) of the compressor turbine vane ring until 100% of the lapped surface is achieved.

(5) Lap the vane ring lockplate contact face (Ref. Fig. 210) until 100% of the lapped surface is achieved.

(6) After lapping, each part must be index marked to its mating part.

(7) Thoroughly clean all lapped surfaces using petroleum solvent (PWC11-027) or (PWC11-031), making sure all traces of compound are removed, then allow to dry.

(8) When the surfaces cannot be restored by lapping, return parts to an overhaul facility for repair.

8. Fits and Clearances

A. General

The following provide the fits and clearances, torques, spring pressures, special assembly, and backlash checks where applicable.

The REF. NO. should be used in conjunction with Figure 211.

B. Dimensional Checks

The DIMENSIONS FOR REF. column indicates minimum and maximum manufacturing dimensions of two mating parts. These dimensions are provided for information only.

The LIMITS column indicates the desired minimum and maximum fits and clearances between new parts, and also the allowable limit to which these parts may wear before replacement is necessary.

The letter T indicates a tight fit; L or no letter, a loose fit.

BY SELECTION means parts must be matched by selection to provide a required fit.

FIT TO means a fitting operation may be required at assembly to obtain the required fit.

An asterisk (*) indicates part(s) should be replaced if any looseness is evident.

Unless otherwise stated, all fits are diametrical. Spline fits are calculated from chordal dimensions.

C. Torque Limits

Unless otherwise stated, apply thread lubricant to all parts that are to be torque-loaded.

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Use engine lubricating oil or equivalent, unless otherwise specified.

Where torque limits for castle nuts are provided in the MINIMUM column only, these nuts should be tightened to the designated torque and then further tightened, if necessary, to properly align the locking slots and holes.

Where MINIMUM and MAXIMUM values are given, the alignment of the locking slot must be obtained without loosening the nut and without exceeding the maximum limit. If this is not possible, back-off the nut half a turn, then retighten. If alignment cannot be accomplished, select another nut.

Torque requirements for interference fit applications, such as studs, may be obtained with or without lubrication, unless otherwise specified.

Table 203 Fits and Clearances

REF. No.	Name	Dim. for Ref. inches		Limits inches		
		Min.	Max.	Min.	Max.	Replace
260	Plate, Lock, C.T. Vane Ring	5.3090	5.3130	0.0050	0.0150	
	Cover, No. 2 Bearing Housing	5.2980	5.3040			
263	Cover, C.T. Bearing Housing	5.3020	5.3040	0.0040T	0.0010	
	Case, Gas Generator Assembly	5.3000	5.3030			
263	Cover, C.T. Bearing Housing	5.3020	5.3040	0.0045T	0.0015	
	Case, Gas Generator Assembly	5.2995	5.3035			
318	Compressor Turbine Blade Radial Tip Clearance			0.011 0.018 (Run Segments) (Ref. NOTE) 0.013 0.016 (New Segments) (Ref. NOTE)		
359	Vane Ring Class, Power Turbine			12.7	13.9	
484	Duct, Combustion Chamber Exit Large	15.945	15.955	0.020	0.040	
	Liner, Combustion Chamber	15.975	15.985			
489	Vane Ring, Turbine	0.1360	0.1400	0.0020	0.0100 (End Float)	
	Cover, No. 2 Bearing Housing	0.1420	0.1460			

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REF. No.	Name	Dim. for Ref. inches		Limits inches		
		Min.	Max.	Min.	Max.	Replace
509	Compressor Turbine, Rotor Diameter			8.546	8.556	8.512
541	Compressor Turbine Stubshaft Center Bolt (Tighten to 600 - 650 lb.in., release to zero, re-tighten to final limits.)			420 460 (Pre-SB1439) 500 550 (Post-SB1439)		
654	Minimum thickness of shroud segment after grinding is 0.055			0.055	0.040	
NOTE: New segment limits are to be used when a new set of segments is required to be ground under field conditions.						

For Aluminum Enamel Coated Gas Generator CaseFor Diffused Aluminide Coated Gas Generator Case Only:Post-SB1434

Table 204 Torque and Stretch

REF. No.	Name	Limits	
		Min.	Max.
NOT APPLICABLE			

Table 205 Spring Pressures

REF. No.	Name	Limits	
		Min.	Max.
NOT APPLICABLE			

Table 206 Special Assembly Procedures

REF. No.	Name	Limits	
		Min.	Max.
	Compressor Turbine Radial Tip Clearance (1) With the CT disk and CT vane removed, measure the CT disk OD and the CT vane assembly shroud ID. Calculate clearances must be within limits. If not, grind or		

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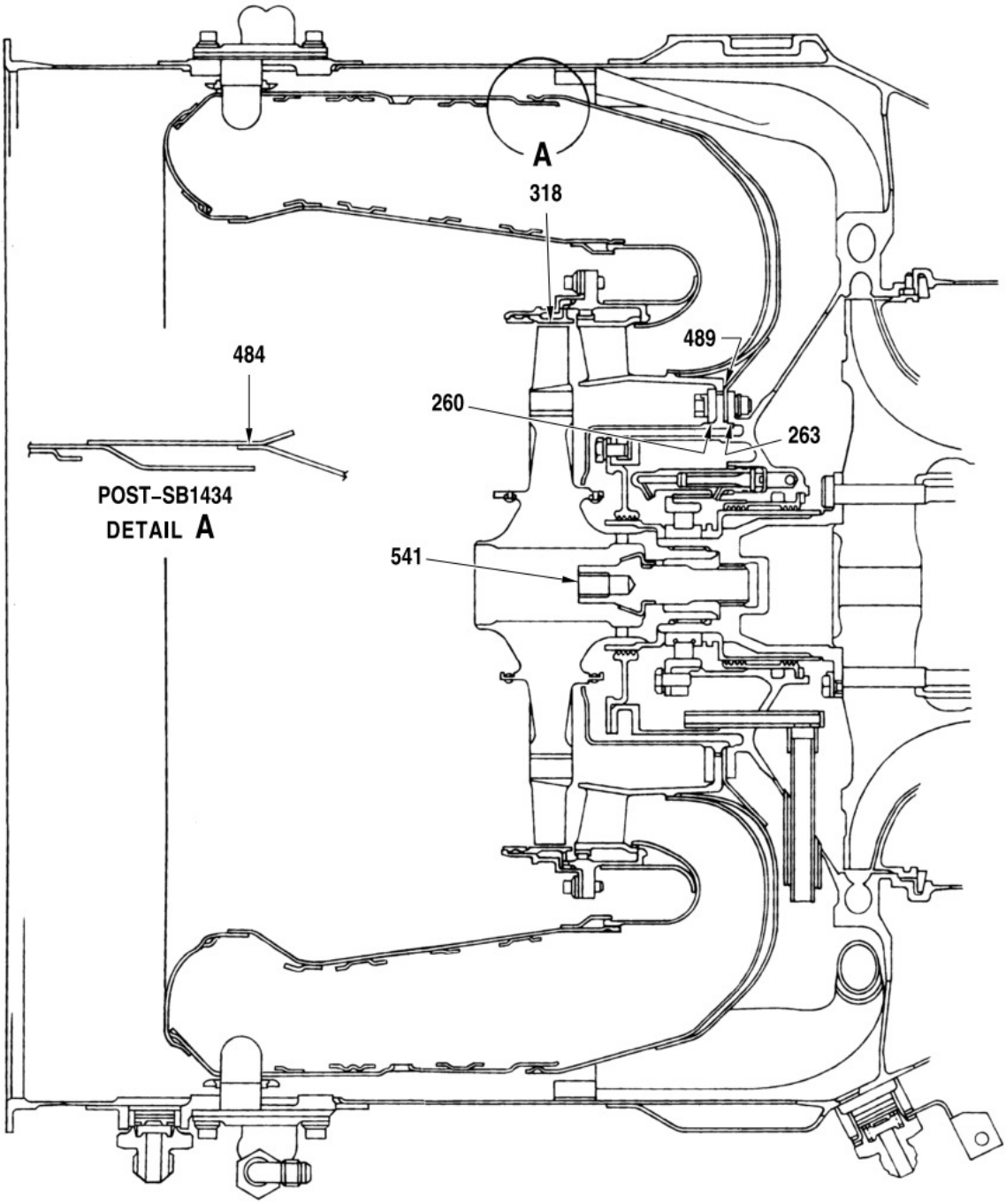
REF. No.	Name	Limits	
		Min.	Max.
318	replace the shrouds. **NOTE**: CT blades may be flat on the tip or may have a taper. For CT blades that are flat on the tip, measure the CT disk OD at the highest point of the blades (leading or trailing edge). For blades with a taper, measure the CT disk OD at the blade trailing edge. (2) With the CT vane assembly mounted in the gas generator, measure the shroud ID at four places, approximately mid segment position. The ID of shroud segments should be within 0.003 (new segments) or 0.005 (run segments) F.I.R. If necessary, machine the shrouds to get the correct clearance. (3) With the CT disk installed, measure radial tip clearance, with wire or tapered feeler gages, at several locations (Ref. Step (1)). Load disc at 180 degrees opposite of the clearance being checked and correct results for bearing play. Machine shroud segments or replace it if necessary to produce clearances within final build limits.		

Table 207 Backlash

REF. No.	Name	Limits	
		Min.	Max.
NOT APPLICABLE			

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Figure 211 Fits and Clearances



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